Claude Opus 4.6 — Comprehensive Technical Analysis

> **Disclaimer:** Claude Opus 4.6 is proprietary. Anthropic has not published exact architecture specs, parameter counts, or training details. Numbers in this document are **speculative estimates** based on publicly available information about comparable frontier models, Anthropic's system card (Feb 2026), official announcements, and independent analysis. Where facts are confirmed, they are cited; where speculative, they are marked as such.

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1. Overview

Claude Opus 4.6 is Anthropic's latest "frontier" model, released **February 5, 2026**, with massive scale and capabilities:

Attribute	Value
Release Date	February 5, 2026
Context Window	1,000,000 tokens (beta)
Architecture	Mixture-of-Experts (MoE) Transformer (speculated)
Total Parameters	$\sim 2\text{–}5$ trillion (speculative)

Active Parameters/Token	$\sim 120\text{–}300$ billion
API Pricing	$5 / $25 per million tokens (input / output)
Open Weights	No — proprietary, API-only
Knowledge Cutoff	~May 2025 (general), ~Aug 2025 (post-fine-tuning)
Safety Level	ASL-3 (Anthropic's classification)

Key Formula — Weight Size

$$\text{Size (bytes)} = N_{\text{params}} \times B_{\text{bytes/param}}$$

Precision	Bytes/Param	Example (1B params)
FP32	4	~ 4 GB
FP16 / BF16	2	~ 2 GB
INT8	1	~ 1 GB
INT4	0.5	~ 0.5 GB

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2. Architecture — Mixture-of-Experts Transformer

Claude Opus 4.6 almost certainly uses **MoE layers**. In MoE transformers, each layer has many feed-forward "expert" modules, but only a small subset are routed for each token.

How MoE Works

- **Sparse activation:** Only $2\text{–}4$ experts (out of $64\text{–}128$) process each token
- **Scaling:** MoE scales capacity without linearly scaling FLOPs
- **Expert Gating:** A learned gating network $G(x)$ routes tokens to relevant experts:

$$y = \sum_{i=1}^{k} G(x)_i \cdot E_i(x)$$

where $G(x)_i$ is the gating weight for expert i, $E_i(x)$ is the expert output, and k is the number of active experts per token.

Speculated Architecture Parameters

Component	Estimated Value
Layers (L)	~ 160
Model Dimension (d_{model})	$\sim 16{,}384\text{–}32{,}768$
Attention Heads (n_h)	~ 128
KV Heads (GQA) (n_{kv})	~ 16
Head Dimension (d_h)	~ 128
FFN Dimension (d_{ff})	$\sim 4 \times d_{\text{model}}$
Number of Experts (E)	$\sim 64\text{–}128$
Active Experts/Token (k)	$\sim 2\text{–}4$
Expert Size	$\sim 10\text{–}20$B params each

Comparison: MoE Models

Model	Total Params	Active Params	Experts	Performance
Mixtral 8×7B	47B	13B	8	Matches Llama-2-70B
DBRX	132B	36B	16	Competitive with GPT-3.5
Llama 4 Behemoth	~ 2T	~ 288B	Many	Frontier-class
Claude Opus 4.6	$\sim 2\text{–}5$**T**	$\sim 120\text{–}300$**B**	$\sim 64\text{–}128$	**SOTA on many benchmarks**

MoE Layout (2T total model estimate)

```
┌──────────────────────────────────────────────────┐
│        Claude Opus 4.6 — MoE Layout              │
│              (2T total params)                   │
├──────────────────────────────────────────────────┤
│                                                  │
│  Shared Components (~200–400B params)            │
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│  ├─ Embedding layers:     ~50–100B          │
│  ├─ Attention (all layers): ~100–200B       │
│  ├─ LayerNorms / biases:    ~1–5B           │
│  └─ Output head:            ~50–100B        │
│                                             │
│  Expert FFNs (~1.6–1.8T params total)       │
│  ├─ 128 experts × ~12–14B each             │
│  ├─ Router/gating networks: ~1–5B           │
│  └─ Only 2–4 experts active per token       │
│                                             │
│  Active params per token: ~120–300B         │
│                                             │
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Attention Computation

Multi-head attention with grouped-query attention (GQA):

$$\text{Attention}(Q, K, V) = \text{softmax}!\left(\frac{Q K^\top}{\sqrt{d_h}}\right) V$$

where $Q \in \mathbb{R}^{n \times d_h}$, $K, V \in \mathbb{R}^{m \times d_h}$.

Total attention parameters per layer:

$$P_{\text{attn}} = d_{\text{model}} \times (n_h \cdot d_h + 2 \cdot n_{kv} \cdot d_h + n_h \cdot d_h)$$

Total FFN parameters per expert (SwiGLU):

$$P_{\text{ffn}} = 3 \times d_{\text{model}} \times d_{ff}$$

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3. Scale & Parameter Estimates

Raw Weight Size Calculations

$$\text{Size} = N_{\text{params}} \times B_{\text{bytes/param}}$$

Precision	Bytes/Param	2T Params	3.5T Params	5T Params
FP32	4	8.0 TB	14.0 TB	20.0 TB
BF16 / FP16	2	4.0 TB	7.0 TB	10.0 TB

FP8	1	2.0 TB	3.5 TB	5.0 TB
INT8	1	2.0 TB	3.5 TB	5.0 TB
INT4	0.5	1.0 TB	1.75 TB	2.5 TB
INT3	0.375	0.75 TB	1.31 TB	1.875 TB
INT2	0.25	0.5 TB	0.875 TB	1.25 TB

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4. Context Window & Long-Range Processing

Opus 4.6's 1M-token context is a major advance. Standard transformers scale **quadratically** with context length:

$$\text{Attention complexity} = \mathcal{O}(n^2 \cdot d_h)$$

KV Cache Memory Formula

$$M_{kv} = 2 \times L \times n_{kv} \times d_h \times S \times b$$

where L = layers, n_{kv} = KV heads, d_h = head dimension, S = sequence length, b = bytes per element.

Example: For the speculated architecture with $S = 1{,}000{,}000$:

$$M_{kv} = 2 \times 160 \times 16 \times 128 \times 1{,}000{,}000 \times 2 \approx 1.25 \text{ TB}$$

Long-Context Techniques

Technique	How It Works
Sparse / Top-k Attention	Retrieve only most relevant keys per query
Infini-attention	Bounded-size compressive memory: $M_{\text{compress}} = \mathcal{O}(d_h^2)$
RoPE Scaling	Extend rotary embeddings: $\theta_i' = \theta_i / s$ where $s > 1$
Compaction	Summarize prior context with smaller model

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5. Performance & Benchmarks

Benchmark	Score	Notes
BigLaw Bench (legal)	90.2%	Highest of any Claude model; 40% perfect scores
Terminal-Bench 2.0 (agentic CLI)	65.4%	#1 among all models
SWE-bench Verified (coding)	$\sim$80.8%	Tied for industry leading
OSWorld-Verified (multi-step SW)	72.7%	—
GPQA-Diamond (physics)	91.3%	Beats GPT-4(o)
ARC-AGI-2 (abstract reasoning)	68.8%	Far above GPT-4/Gemini
MMLU (multitask knowledge)	91.1%	10-choice format
GDPval-AA (economics)	1606 Elo	$+144$ over GPT-5.2 ($\sim$1462)
Humanity's Last Exam	#1	Advanced reasoning & synthesis

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6. Training Data

Chinchilla Scaling Law (Hoffmann et al., 2022)

The compute-optimal training rule:

$$D_{\text{optimal}} \approx 20 \times N$$

where D = training tokens and N = model parameters. For a 2T-param model:

$$D_{\text{optimal}} \approx 20 \times 2 \times 10^{12} = 4 \times 10^{13} \text{ tokens} = 40\text{T tokens}$$

Industry Comparisons

Model	Params (N)	Training Tokens (D)	D/N Ratio
Chinchilla	70B	1.4T	20×
GPT-4	$\sim$1.8T	$\sim$13T	$\sim$7×
Llama 3.1 405B	405B	>15T	$\sim$37×
Llama 4 Behemoth	$\sim$2T	>30T	$\sim$15×
Claude Opus 4.6	$\sim 2\text{–}5$**T**	**Unknown**	**Est. 20–40T+**

Disclosed Data Sources (System Card)

- Broadly-crawled web text (up to May 2025)
- Licensed corpora and books
- Contractor-curated/annotated data
- Opted-in Claude user content
- Synthetic/self-generated data

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7. Training & Inference Infrastructure

Training (Estimated)

Aspect	Estimate
GPU Count	20,000–60,000 NVIDIA H100
Training Duration	3–6 months
Training Data	20–40+ trillion tokens
Interconnect	NVLink / NVSwitch
Estimated FLOPs	$\sim 3.6 \times 10^{25}$

Training FLOPs Estimate (Kaplan approximation)

$$C \approx 6 \times N \times D$$

For $N = 2 \times 10^{12}$ and $D = 30 \times 10^{12}$:

$$C \approx 6 \times 2 \times 10^{12} \times 30 \times 10^{12} = 3.6 \times 10^{26} \text{ FLOPs}$$

Inference — GPU RAM Requirements

Component	Params	BF16 Size	In GPU RAM?
Shared attention + embeddings	$\sim$300B	$\sim$600 GB	Yes (always)
Each expert FFN	$\sim$14B	$\sim$28 GB	Only if routed
All 128 experts	$\sim$1.7T	$\sim$3.4 TB	No — offload
Active experts (2–4)	$\sim$28–56B	$\sim$56–112 GB	Yes
Minimum GPU RAM	**$\sim$330–360B**	**$\sim$660–720 GB**	—

Hardware Configurations

Setup	Total GPU RAM	Feasibility
$8\times$ H100 80GB	640 GB	Tight, needs expert offloading
$16\times$ H100 80GB	1.28 TB	Comfortable

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8. Open-Weights Status

Claude Opus 4.6 is **not open-sourced**:

- Available only via API ($5/$25 per million tokens)
- Weights preserved internally "at minimum for the lifetime of Anthropic"
- RSP v3.0 (Feb 2026) strengthens safeguards against weight theft
- No credible leaks or unauthorized weights have surfaced
- Contrast: Meta (Llama), xAI (Grok), Alibaba (Qwen) release open weights

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9. Hypothetical Open-Weight Release

> **Caveat:** This section is entirely speculative.

Sharding Formula

$$N_{\text{shards}} = \left\lceil \frac{\text{Total Size}}{\text{Shard Size}} \right\rceil$$

Shard Size	Shards (2T @ BF16)	Shards (5T @ BF16)
5 GB	$\sim$800 files	$\sim$2,000 files
10 GB	$\sim$400 files	$\sim$1,000 files
20 GB	$\sim$200 files	$\sim$500 files

GGUF Community Quantizations

Quant Method	Bits/Param	Size (2T)	Size (5T)	Quality
Q8_0	$\sim$8.5	$\sim$2.1 TB	$\sim$5.3 TB	Near-lossless
Q6_K	$\sim$6.6	$\sim$1.6 TB	$\sim$4.1 TB	Excellent
Q5_K_M	$\sim$5.7	$\sim$1.4 TB	$\sim$3.6 TB	Very good
Q4_K_M	$\sim$4.8	$\sim$1.2 TB	$\sim$3.0 TB	Good (most popular)
Q3_K_M	$\sim$3.9	$\sim$0.98 TB	$\sim$2.4 TB	Acceptable
Q2_K	$\sim$3.2	$\sim$0.8 TB	$\sim$2.0 TB	Degraded
IQ2_XXS	$\sim$2.1	$\sim$0.5 TB	$\sim$1.3 TB	Research-grade

GPU Quantization Formats

Format	Typical Bits	Size (2T)	Best For

GPTQ	4-bit	$\sim$1.0–2.0 TB	GPU inference (AutoGPTQ)
AWQ	4-bit	$\sim$1.0–1.2 TB	GPU inference (vLLM, TGI)
EXL2	2–6 bpw	$\sim$0.5–1.5 TB	ExLlamaV2
HQQ	2–4 bit	$\sim$0.5–1.0 TB	Half-Quadratic Quantization
AQLM	2-bit	$\sim$0.5 TB	Extreme compression

Download Summary

Scenario	Format	Size (2T)	Size (5T)
Full official	SafeTensors BF16	$\sim$4.0 TB	$\sim$10 TB
Best quality quant	GGUF Q8_0	$\sim$2.1 TB	$\sim$5.3 TB
Best tradeoff	GGUF Q4_K_M	$\sim$1.2 TB	$\sim$3.0 TB
Minimum viable	GGUF Q2_K	$\sim$0.5–0.8 TB	$\sim$1.3–2.0 TB
GPU-optimized	AWQ 4-bit	$\sim$1.0 TB	$\sim$2.5 TB

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10. Multimodality — Vision & Computer Use

Vision Encoder Architecture

$$\text{Image} \xrightarrow{\text{Patch}} \text{ViT} \xrightarrow{\text{Project}} \text{LLM embedding space}$$

Patch tokenization formula:

$$N_{\text{visual_tokens}} = \frac{H}{P} \times \frac{W}{P}$$

where H, W = image dimensions, P = patch size (typically 14 or 16 pixels).

Image Resolution	Patch Size	Visual Tokens
224×224	14	256
336×336	14	576
672×672	14	2,304
1344×1344	14	9,216

Vision Encoder Parameters

Component	Params	Size (BF16)
ViT-Large (ViT-L/14)	$\sim$307M	$\sim$614 MB
ViT-Huge (ViT-H/14)	$\sim$632M	$\sim$1.26 GB
ViT-Giant (ViT-G/14)	$\sim$1.8B	$\sim$3.6 GB
Cross-attention projector	$\sim$100–500M	$\sim$0.2–1.0 GB
Total	**$\sim$1–3B**	**$\sim$2–6 GB ($< 0.1%$ of total)**

Visual Token Context Cost

$$\text{Effective text context} = 1{,}000{,}000 - N_{\text{images}} \times N_{\text{visual_tokens/image}}$$

Computer Use Pipeline

$$\text{Screenshot} \xrightarrow{\text{ViT}} \text{Visual tokens} \xrightarrow{\text{LLM}} \text{Action}(x, y, \text{click/type/scroll})$$

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11. Sampling & Decoding Strategies

Temperature Scaling

$$P(t_i) = \frac{\exp(z_i / T)}{\sum_j \exp(z_j / T)}$$

where z_i = logit for token i, T = temperature.

Temperature	Effect	Use Case
$T = 0$	Greedy (argmax)	Deterministic code
$T = 0.3$	Low randomness	Professional writing
$T = 0.7$	Balanced (default)	General conversation
$T = 1.0$	Full softmax	Creative writing
$T > 1.0$	High randomness	Brainstorming

Top-p (Nucleus) Sampling

Select smallest set V_p such that:

$$\sum_{t_i \in V_p} P(t_i) \geq p$$

Then renormalize and sample from V_p only.

Repetition Penalty

$$z'_i = \begin{cases} z_i / \alpha & \text{if } z_i > 0 \text{ and } t_i \in \text{context} \ z_i \times \alpha & \text{if } z_i \leq 0 \text{ and } t_i \in \text{context} \end{cases}$$

where $\alpha > 1$ penalizes already-seen tokens.

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12. Prompt Caching

KV Cache Reuse Formula

$$M_{\text{cached}} = 2 \times L \times n_{kv} \times d_h \times S_{\text{prefix}} \times b$$

Cost Savings

Operation	Cost (Opus 4.6)
Input (uncached)	$5.00 / M tokens
Cache write	$6.25 / M tokens ($1.25\times$)
Cache read (hit)	$0.50 / M tokens ($0.1\times$)
Output	$25.00 / M tokens

Example: 10,000-token system prompt used 1,000 times:

$$\text{Without cache: } 1{,}000 \times 10{,}000 \times $5/\text{M} = $50.00$$

$$\text{With cache: } $0.0625 + 1{,}000 \times 10{,}000 \times $0.50/\text{M} = $5.06$$

Savings: $\sim 90%$

Cache Storage Size (10K tokens)

$$M_{\text{cache}} = 2 \times 160 \times 16 \times 128 \times 10{,}000 \times 2 = 12.5 \text{ GB}$$

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13. Normalization — RMSNorm vs LayerNorm

LayerNorm (Original Transformer)

$$\text{LayerNorm}(x) = \gamma \cdot \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}} + \beta$$

where $\mu = \frac{1}{d}\sum x_i$, $\sigma^2 = \frac{1}{d}\sum(x_i - \mu)^2$. Parameters: $2 \times d_{\text{model}}$.

RMSNorm (Likely Used by Opus 4.6)

$$\text{RMSNorm}(x) = \gamma \cdot \frac{x}{\sqrt{\frac{1}{d}\sum x_i^2 + \epsilon}}$$

Parameters: d_{model} (scale γ only, no shift).

- $\sim 10\text{–}15\%$ faster than LayerNorm
- Empirically equivalent quality

Total RMSNorm parameters:

$$P_{\text{norm}} = 2 \times L \times d_{\text{model}} = 2 \times 160 \times 16{,}384 = 5.24\text{M} \quad (\text{negligible})$$

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14. Activation Functions — SwiGLU

SwiGLU (Standard in Modern Transformers)

$$\text{SwiGLU}(x) = \left(\text{Swish}(xW_{\text{gate}}) \odot xW_{\text{up}}\right) W_{\text{down}}$$

where:

$$\text{Swish}(x) = x \cdot \sigma(\beta x)$$

and σ is the sigmoid function. Requires **three** weight matrices per FFN:

$$P_{\text{ffn}} = 3 \times d_{\text{model}} \times d_{ff}$$

Why SwiGLU over ReLU:

- $\sim 1\text{–}2%$ better perplexity at same compute
- Smoother gradients → more stable training
- Multiplicative gating provides richer expressivity

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15. Learning Rate Schedule

Warmup + Cosine Decay

$$\eta(t) = \begin{cases} \eta_{\max} \cdot \dfrac{t}{T_{\text{warmup}}} & t \leq T_{\text{warmup}} \[10pt] \eta_{\min} + \dfrac{1}{2}(\eta_{\max} - \eta_{\min})\left(1 + \cos!\left(\dfrac{\pi(t - T_{\text{warmup}})}{T_{\text{total}} - T_{\text{warmup}}}\right)\right) & t > T_{\text{warmup}} \end{cases}$$

Typical Hyperparameters

Parameter	Typical Value
Peak LR ($\eta_{\max}$)	1×10^{-4} to 3×10^{-4}
Final LR ($\eta_{\min}$)	$\eta_{\max}/10$ to $\eta_{\max}/100$
Warmup steps (T_{warmup})	2,000–5,000
β_1, β_2 (Adam)	0.9, 0.95
Weight decay	0.1
Gradient clipping	1.0 (max grad norm)
Batch size	2–16M tokens

Batch Size Scaling

$$B_{\text{total}} = B_{\text{micro}} \times N_{\text{accum}} \times N_{\text{data_parallel}}$$

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16. Loss Functions

Primary Pre-training Loss (Cross-Entropy)

$$\mathcal{L}_{\text{LM}} = -\frac{1}{T}\sum_{t=1}^{T} \log P_\theta(x_t \mid x_{<t})$$

MoE Auxiliary Losses

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{LM}} + \alpha \cdot \mathcal{L}_{\text{balance}} + \beta \cdot \mathcal{L}_z$$

Loss	Purpose	Typical Weight
$\mathcal{L}_{\text{LM}}$	Language modeling	1.0
$\mathcal{L}_{\text{balance}}$	Prevent expert collapse	0.01–0.1
$\mathcal{L}_z$	Router logit stability	0.001

Load Balancing Loss

$$\mathcal{L}_{\text{balance}} = E \cdot \sum_{i=1}^{E} f_i \cdot p_i$$

where f_i = fraction of tokens routed to expert i, p_i = average gating probability for expert i.

Perplexity

$$\text{PPL} = \exp(\mathcal{L}_{\text{LM}})$$

Frontier models typically achieve $\text{PPL} \approx 5\text{–}8$ on standard benchmarks.

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17. Numerical Stability & Mixed Precision

Format Comparison

Format	Exponent	Mantissa	Range	Precision
FP32	8 bits	23 bits	$\pm 3.4 \times 10^{38}$	$\sim$7 digits

FP16	5 bits	10 bits	$\pm 65{,}504$	$\sim$3.3 digits
BF16	8 bits	7 bits	$\pm 3.4 \times 10^{38}$	$\sim$2.4 digits
FP8 (E4M3)	4 bits	3 bits	± 448	$\sim$1.5 digits
FP8 (E5M2)	5 bits	2 bits	$\pm 57{,}344$	$\sim$1.2 digits

Loss Scaling (for FP16 training)

$$\hat{\mathcal{L}} = s \cdot \mathcal{L}, \qquad \hat{g} = \frac{g}{s}$$

BF16 typically doesn't need this — hence it is preferred.

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18. Hardware Failure & Checkpointing

Failure Rate at Scale

With $N = 32{,}000$ GPUs, $\sim 1\%$ annual failure rate:

$$p_{\text{fail/GPU/day}} \approx \frac{0.01}{365} \approx 2.7 \times 10^{-5}$$

$$\mathbb{E}[\text{failures/day}] = 32{,}000 \times 2.7 \times 10^{-5} \approx 0.87$$

Over 60 days: ~ 52 GPU failures expected.

Checkpointing Cost

Full model state (weights $+$ optimizer $+$ RNG):

$$M_{\text{state}} = N_{\text{params}} \times 16 \text{ bytes} = 2\text{T} \times 16 = 32 \text{ TB}$$

$$T_{\text{checkpoint}} = \frac{32{,}000 \text{ GB}}{100 \text{ GB/s}} \approx 320 \text{ s} \approx 5.3 \text{ min}$$

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19. Continuous Batching & Inference Scheduling

GPU Utilization

$$\text{Utilization}_{\text{continuous}} = \frac{\sum_i T_i^{\text{compute}}}{B \times T_{\max}} \approx 2\text{–}3\times \text{ vs static}$$

Iteration-Level Scheduling

```
Step 1: [User A token 1,  User B token 45, User C token 200, ...]
Step 2: [User A token 2,  User B token 46, User D token 1 (new!), ...]
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20. Distillation & Model Compression

Knowledge Distillation Loss

$$\mathcal{L}_{\text{distill}} = (1 - \alpha)\mathcal{L}_{\text{CE}}(y, \hat{y}) + \alpha T^2 \cdot \text{KL}\left(P_{\text{teacher}}^T \,\|\, P_{\text{student}}^T\right)$$

where T = temperature for softening, α = interpolation weight.

Model	Speculated Size	Distilled From
Opus 4.6	$2\text{–}5$T total	Pre-trained from scratch
Sonnet 4.6	$\sim$200–500B?	Likely distilled from Opus
Haiku 4.6	$\sim$30–70B?	Likely distilled from Sonnet/Opus

Structured Pruning

$$W_{\text{pruned}} = W \odot M, \quad M_{ij} = \mathbf{1}[|W_{ij}| > \theta]$$

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21. Expert Specialization Analysis

Expert Utilization Entropy

$$H_{\text{expert}} = -\sum_{i=1}^{E} p_i \log p_i$$

- $H = \log E$ → perfectly balanced
- $H \ll \log E$ → some experts dominate (collapse risk)

Expert Correlation Matrix

$$C_{ij} = \text{Corr}!\left(\mathbf{1}[\text{expert } i \text{ active}],; \mathbf{1}[\text{expert } j \text{ active}]\right)$$

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22. Structured Output / JSON Mode

Constrained Decoding

$$P'(t_i) = \begin{cases} P(t_i) / Z & \text{if } t_i \text{ is valid given grammar state} \ 0 & \text{otherwise} \end{cases}$$

where $Z = \sum_{j \in \text{valid}} P(t_j)$ is the normalizing constant.

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23. Watermarking

Statistical Watermarking (Kirchenbauer et al., 2023)

At each step, partition vocabulary using $h_t = \text{Hash}(t_{i-1})$:

$$\text{Green list} = {t : h_t(t) < |V|/2}$$

$$z'_i = z_i + \delta \quad \text{if } t_i \in \text{green list}$$

Detection:

$$z\text{-score} = \frac{|G| - T/2}{\sqrt{T/4}}$$

If $z > 4$, almost certainly watermarked.

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24. Logprobs & Uncertainty

Log-Probabilities

$$\text{logprob}(t_i) = \log P(t_i \mid x_{<i})$$

Entropy as Uncertainty

$$H(t) = -\sum_i P(t_i \mid x_{<t}) \log P(t_i \mid x_{<t})$$

Expected Calibration Error

$$\text{ECE} = \sum_{b=1}^{B} \frac{n_b}{N} \left|\text{acc}(b) - \text{conf}(b)\right|$$

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25. Tool Use / Function Calling

Agent Loop — Token Cost

Input grows each turn (all previous messages re-sent):

$$C_{\text{input}}(i) = C_{\text{system}} + \sum_{j=1}^{i-1} \left(C_{\text{output}}(j) + C_{\text{tool_result}}(j)\right)$$

$$C_{\text{agent}} = \sum_{i=1}^{N_{\text{turns}}} \left(C_{\text{input}}(i) + C_{\text{output}}(i)\right)$$

Example: 20-turn agent loop, 50K-token conversations:

$$\text{Without caching: } \approx 20 \times 50{,}000 \times \$5/\text{M} + 20 \times 2{,}000 \times \$25/\text{M} = \$6.00$$

$$\text{With caching: } \approx \$1.50 \quad (70\% \text{ savings})$$

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26. Multilingual Analysis

Tokenizer Fertility

$$F_{\text{lang}} = \frac{N_{\text{tokens}}}{N_{\text{characters or words}}}$$

Higher fertility $=$ less efficient $=$ more expensive per unit of meaning.

Language	Tokens/Word	Effective Context (1M tokens)
English	$\sim$1.3	$\sim$750K words
Spanish	$\sim$1.5	$\sim$667K words

Chinese	$\sim$1.5–2.0/char	$\sim$500K–667K chars
Japanese	$\sim$2.0–3.0/char	$\sim$333K–500K chars
Arabic	$\sim$2.0	$\sim$500K words
Hindi	$\sim$3.0–4.0	$\sim$250K–333K words

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27. Curriculum Learning & Data Scheduling

Data Mix Evolution

Phase	% Compute	Data Emphasis
Phase 1 (0–60%)	60%	Broad web text
Phase 2 (60–85%)	25%	Higher-quality (Wikipedia, books)
Phase 3 (85–95%)	10%	Code, math, reasoning
Phase 4 (95–100%)	5%	Instruction-following

Annealing

$$\eta_{\text{anneal}} = \eta_{\min} + (\eta_{\text{current}} - \eta_{\min}) \cdot \cos!\left(\frac{\pi t}{2 T_{\text{anneal}}}\right)$$

---

28. Benchmark Contamination Testing

N-gram Overlap

$$\text{Contamination}(B) = \frac{|{x \in B : \exists, d \in D,; \text{ngram_overlap}(x, d) > \theta}|}{|B|}$$

Detection Methods

Method	Description
N-gram overlap	Check for verbatim matches in training data
Canary strings	Plant unique strings; test if model completes them

Rephrased evaluation	Large drops = memorization, not understanding

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29. Interpretability / Mechanistic Interpretability

Sparse Autoencoders (SAEs)

$$h = \text{ReLU}(W_{\text{enc}} \cdot x + b_{\text{enc}})$$

$$\hat{x} = W_{\text{dec}} \cdot h + b_{\text{dec}}$$

$$\mathcal{L}_{\text{SAE}} = |x - \hat{x}|^2 + \lambda |h|_1$$

Anthropic's published work:

- Mapped **millions of features** in Claude 3 Sonnet (May 2024)
- Found features for cities, code languages, emotions, safety concepts
- Can **steer behavior** by amplifying/suppressing features

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30. Streaming Architecture

Latency Breakdown

$$T_{\text{total}} = T_{\text{TTFT}} + N_{\text{output}} \times T_{\text{per_token}} + T_{\text{network}}$$

Phase	Typical Latency
TTFT (short prompt)	0.5–2s
TTFT (100K context)	5–15s
TTFT (1M context)	30–120s
Per-token generation	15–30ms
500-token response	8–17s

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31. Thinking Guardrails & Adaptive Thinking

Effort Levels

Level	Behavior
`low`	Minimal thinking, fast/cheap
`medium`	Balanced
`high` (default)	Deep, selective extended thinking
`max`	Maximum depth, revisits, caution

Speculative Internal Model (Not Official)

$$T = f(C \times E)$$

$$D \approx k \times C \times E$$

where C = task complexity, $E \in {1, 2, 3, 4}$ = effort level, T = trigger probability, D = thinking depth.

Safety Integration

- $0%$ attack success on agentic coding attacks
- Constitutional AI checks during thinking
- Compaction at $\sim 5%$ of cases

---

32. Version History & Model Lineage

```
Claude 1.0 (Mar 2023)
 └─ Claude 2.0 (Jul 2023)
     └─ Claude 3 (Mar 2024): Haiku / Sonnet / Opus
         └─ Claude 3.5 Sonnet (Jun 2024)
             └─ Claude 4.6 family (Feb 2026)
                 ├─ Opus 4.6    ← THIS
                 ├─ Sonnet 4.6
                 └─ Haiku 4.6 (?)
```

Feature	Claude 3 Opus	Claude 4.6 Opus

Context	200K	1M (beta)
Architecture	Dense (likely)	MoE (speculated)
Params (est.)	$\sim$200–400B	$\sim$2–5T
Thinking	Extended (optional)	Adaptive (default)
Coding	Good	SOTA (80.8% SWE-bench)

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33. Weight Initialization

Standard:

$$W \sim \mathcal{N}!\left(0, \frac{\sigma}{\sqrt{d_{\text{model}}}}\right)$$

Scaled for Deep Networks ($L > 160$):

$$W_{\text{out}} \sim \mathcal{N}!\left(0, \frac{\sigma}{\sqrt{2L}}\right)$$

MoE Router:

$$W_{\text{router}} \sim \mathcal{N}(0, 0.01)$$

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34. Gradient Accumulation

$$g_{\text{accumulated}} = \frac{1}{K}\sum_{k=1}^{K} g_k$$

$$B_{\text{eff}} = B_{\text{micro}} \times K \times N_{\text{DP}}$$

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35. Safety Classifiers

```
User Input → [Input Classifier] → Model → [Output Classifier] → Response
                    ↓                              ↓
              Block if harmful              Block if harmful
```

- $\sim 1\text{–}10$B params total (negligible vs main model)
- Add $\sim 10\text{–}50$ ms latency per request

36. A/B Testing & Deployment Pipeline

Canary Deployment

$$\text{Response} = \begin{cases} M_{\text{new}} & \text{with probability } p_{\text{canary}} \ M_{\text{old}} & \text{with probability } 1 - p_{\text{canary}} \end{cases}$$

37. Fill-in-the-Middle (FIM) for Code

FIM Training Mix

$$\text{FIM mix} = (1 - r) \times \text{autoregressive} + r \times \text{FIM}$$

Typically $r = 0.5$ (50% of code data).

38. API Rate Limits & Operational Details

Tier	Requests/min	Tokens/min	Tokens/day
Free	5	20,000	300,000
Build	50	40,000	1,000,000
Scale	1,000	400,000	50,000,000
Enterprise	Custom	Custom	Custom

Cost Formula

$$\text{Cost} = \frac{N_{\text{input}} \times P_{\text{input}} + N_{\text{output}} \times P_{\text{output}} + N_{\text{thinking}} \times P_{\text{output}}}{10^6}$$

Scenario	Input	Thinking	Output	Cost
Simple question	100	0	200	$0.0055
Complex reasoning	1,000	10,000	500	$0.268
Agentic (10 turns)	50,000	50,000	5,000	$1.63

Full 1M context	1,000,000	5,000	2,000	$5.18

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39. Economic Analysis

Revenue Estimate (10M API requests/day)

$$\text{Daily revenue} = 10^7 \times (2{,}000 \times $5/\text{M} + 500 \times $25/\text{M}) = $225{,}000/\text{day}$$

$$\text{Annual revenue} \approx $82\text{M}$$

Infrastructure Cost

$$\text{Serving cost} \approx 1{,}000 \times $2.50/\text{hr} \times 8{,}760\text{hr/yr} = $21.9\text{M/yr}$$

Gross margin: $\sim 70\text{–}75%$

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40. Regulatory & Legal Context

EU AI Act

- Opus 4.6 = **General-Purpose AI (GPAI)** model
- Estimated training compute: $\sim 3.6 \times 10^{25}$ FLOPs $>$ 10^{25} threshold → **systemic risk**
- Requirements: Adversarial testing, incident reporting, energy disclosure

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41. Release & Competitive Context

Date	Event
Feb 5, 2026	Opus 4.6 released
Feb 5, 2026	OpenAI releases GPT-5.3-Codex ($\sim$15 min after)
Feb 17, 2026	Claude Sonnet 4.6 released
$\sim$Mar 5, 2026	OpenAI releases GPT-5.4

Notable Achievements

- Solved the **graph decomposition conjecture** (31 explorations, $\sim$1 hour)
- #1 on Arena.ai leaderboard
- Agent teams for enterprise workflows
- Computer use (screenshots → actions)

Security Issues

- Red-teamed in 30 minutes
- Service outage March 2, 2026
- $200M Pentagon contract unraveled

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42. Who Built It

Person	Role
Dario Amodei	CEO & Co-founder
Daniela Amodei	President & Co-founder
Boris Cherny	Head of Claude Code

Built by hundreds of Anthropic employees. 213-page system card credits it as a team effort.

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43. Sources

- Anthropic — Claude Opus 4.6 system card (Feb 2026)
- Anthropic — RSP v3.0 and deprecation commitments
- Hoffmann et al. — "Training Compute-Optimal LLMs" (Chinchilla), 2022
- Jiang et al. — "Mixtral of Experts", 2024
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- Meta — Llama 3.1 and Llama 4 model cards
- Kirchenbauer et al. — "A Watermark for Large Language Models", 2023
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- Leaked GPT-4 architecture analysis (SemiAnalysis)
- Arena.ai leaderboard (March 2026)

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